FATIMA EZZAHRA BOUKHORSSA

Master's Student (M2) — Applied Mathematics & Artificial Intelligence

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Profile

Master's student (M2) in Applied Mathematics for AI, seeking a 6-month end-of-studies internship (starting 2027) in Data Science, Machine Learning, Generative AI or Data Analytics. Passionate about scientific research and about the potential of AI to solve complex, high-impact problems, I build end-to-end data projects: an agentic multi-RAG system evaluated with RAGAS, an ML pipeline on 5M+ records, A/B testing and predictive modeling. Strong foundation in statistics and Python/SQL. Curious and always learning, I welcome any opportunity that will deepen my scientific profile and let me explore new domains.

Technical Skills

- **Data Science & Machine Learning:** Scikit-Learn, SciPy, Classification, Regression (OLS), Random Forest, Feature Engineering, Anomaly Detection, Model Evaluation.
- **Generative AI & RAG:** RAG architectures (Hybrid Search, Re-ranking), Agentic Systems (LangChain, LangGraph), Self-Correction (Reflection), Vector Database (Qdrant), RAGAS.
- **Data Engineering & Big Data:** Python (Pandas, NumPy), SQL (PostgreSQL, MySQL), ETL/ELT Pipelines, Large-scale data processing (5M+ rows), Linux/Bash, Git/GitHub, Docker.
- **Mathematics & Statistics:** Statistical Inference, Hypothesis Testing (A/B Testing, Z-test), Probability, Linear Algebra, Optimization.
- **BI & Visualization:** Power BI, Tableau, Streamlit, Interactive Dashboards, Reporting.

Key Projects

Agentic Multi-RAG System & Automated Evaluation *Personal project | In progress*

- Advanced RAG pipeline: hybrid retrieval (BM25 + Qdrant dense vectors) with cross-encoder re-ranking.
- Self-correction (Reflection) loop to validate the faithfulness of answers.
- Automated evaluation with RAGAS (Faithfulness, Answer Relevancy, Context Precision).
- *Stack:* Python, LangChain, LangGraph, Qdrant, Sentence-Transformers (cross-encoder), RAGAS, Streamlit.

Schengen Border Risk Analysis (Big Data & ML) *Personal project | 2026*

- Processing and structuring of 5M+ Frontex records (2009–2026).
- Predictive classification pipelines to categorize alert levels.
- *Stack:* Python, Pandas, Scikit-Learn, Feature Engineering, Streamlit, Git.

Statistical Evaluation Harness & A/B Testing *Personal project | 2026*

- Automated statistical testing pipeline (Z-test, $\alpha = 5\%$) to validate models and generate performance reports.
- *Stack:* Python, SciPy, Pandas, Inferential Statistics.

ETL Pipeline & Predictive Modeling (Superstore BI) *Personal project | 2026*

- Automated cleaning of $\sim$10k sales orders and predictive margin modeling (OLS, Random Forest), delivered via Power BI / Streamlit.
- *Stack:* Python, SQL, Pandas, Scikit-Learn, Power BI, Streamlit.

Education

Master's in Applied Mathematics for AI (Excellence Track) — Mohammed First University, Nador *2025 – Present*
Bachelor's in Applied Mathematics for AI (Excellence Track) — Mohammed First University, Nador *2024 – 2025*
DUT in Networks & Telecommunications — Higher School of Technology (EST), Fès *2022 – 2024*

Other Experience

- **Technical internship, PLANET SYSTEM (2024):** network administration, firewalls and systems security.
- **IoT project (DUT final project, 2024):** real-time acquisition (Arduino/ESP32) and data streaming to a mobile application.
- **Image encryption (2025):** DCT, permutation and XOR algorithms in MATLAB.

Languages & Interests

Languages: Arabic (native), French (fluent, B2/C1), English (professional, B2/C1), Spanish (basic).
Interests: Chess, Sudoku, Pilates, Yoga, AI technology watch.